DEPARTMENT OF ACADEMIC AFFAIRS

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Project Report

On

# SMART PARKING SYSTEM USING IOT Node MCU WITH RFID AUTHENTICATION

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Discover. Learn. Empower.

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# SMART PARKING SYSTEM USING IOT Node MCU WITH RFID AUTHENTICATION

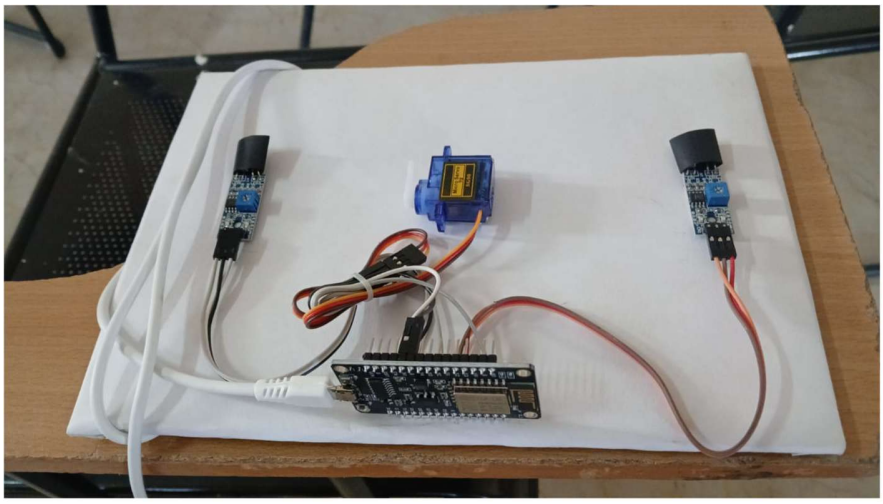
## 1. INTRODUCTION

In today's rapidly expanding urban landscape, the number of vehicles on roads has increased dramatically, leading to severe parking-related challenges. Conventional parking systems rely heavily on manual monitoring and lack proper automation, resulting in inefficient space utilization, increased traffic congestion, longer waiting times, and frequent unauthorized access. These issues not only waste time and fuel but also contribute to environmental pollution and user frustration.

To address these problems, the concept of smart parking systems has emerged as a key solution within modern smart city infrastructure. A smart parking system leverages embedded systems, automation, and sensor technologies to optimize parking operations, enhance security, and provide real-time information to users.

This project presents the design and implementation of a Smart Parking System using Arduino with RFID Authentication, which automates vehicle entry, exit, and parking slot management. The system is built around the Arduino Uno microcontroller, which acts as the central processing unit responsible for coordinating all system operations.

The system integrates multiple hardware components such as an RFID module (RC522), infrared (IR) sensors, a servo motor, and a 16×2 LCD display with an I2C interface. The RFID module ensures that only authorized users can access the parking facility by verifying unique identification cards. IR sensors are strategically placed to detect vehicle movement at entry and exit points and to monitor parking slot occupancy. A servo motor acts as an automated gate barrier, opening or closing based on authentication status and slot availability. The LCD display provides real-time feedback, including system prompts and parking availability information.



## 2. OBJECTIVES

The primary objective of this project is to design and implement an intelligent parking management system using Arduino and embedded technologies. The specific objectives are:

- To design and develop an automated smart parking system using Arduino Uno
- To implement RFID-based authen ca on for secure vehicle access
- To detect vehicle movement using IR sensors for accurate entry and exit tracking
- To monitor and display real- me parking slot availability
- To control gate opera ons automa cally using a servo motor
- To prevent unauthorized access and eliminate over-parking
- To provide a user-friendly interface through an LCD display
- To reduce human interven on and manual errors through automa on
- To improve parking efficiency and op mal space u liza on
- To demonstrate real-world applica on of IoT and embedded systems
- To develop a low-cost and scalable solu on
- To enhance system reliability and response me
- To provide real- me status updates to users
- To create a founda on for future smart city integra ons

## 3. COMPONENTS REQUIRED

### 1. NodeMCU (ESP8266)

The **NodeMCU (ESP8266)** acts as the central controller of the system with built-in WiFi capability. It processes inputs from sensors and communicates data to cloud platforms like Blynk, while also controlling outputs such as servo motors and displays.

Key Features:

- Based on **ESP8266 WIFI microcontroller**
- Built-in **WIFI module** (no extra hardware needed)
- Operates at **3.3V**
- Has **digital I/O pins** with PWM support
- Supports **UART, SPI, and I2C communication**
- Can be programmed using **Arduino IDE**

- Compact and low power consumption

Functions in Project:

- Executes control logic
- Processes sensor data
- Controls servo motor (gate system)
- Sends data to **Blynk/cloud**
- Manages authentication and decision-making
- Enables **remote monitoring via mobile app**

## 2. IR Sensors

Infrared sensors detect the presence of vehicles at entry and exit points.

Working Principle:

- Emits infrared light
- Detects reflected signals from objects
- Sends digital signal to Arduino Role:

- Detect vehicle entry and exit
- Maintain parking count
- Update slot availability

## 3. Servo Motor (SG90)

The servo motor acts as an automatic gate controller.

Working Principle:

- Rotates to specific angles (0° to 180°)

- Controlled using PWM signals Role:

- Opens gate for authorized access
- Closes gate a er vehicle passes

### 4. Jumper Wires

Used for electrical connections between components.

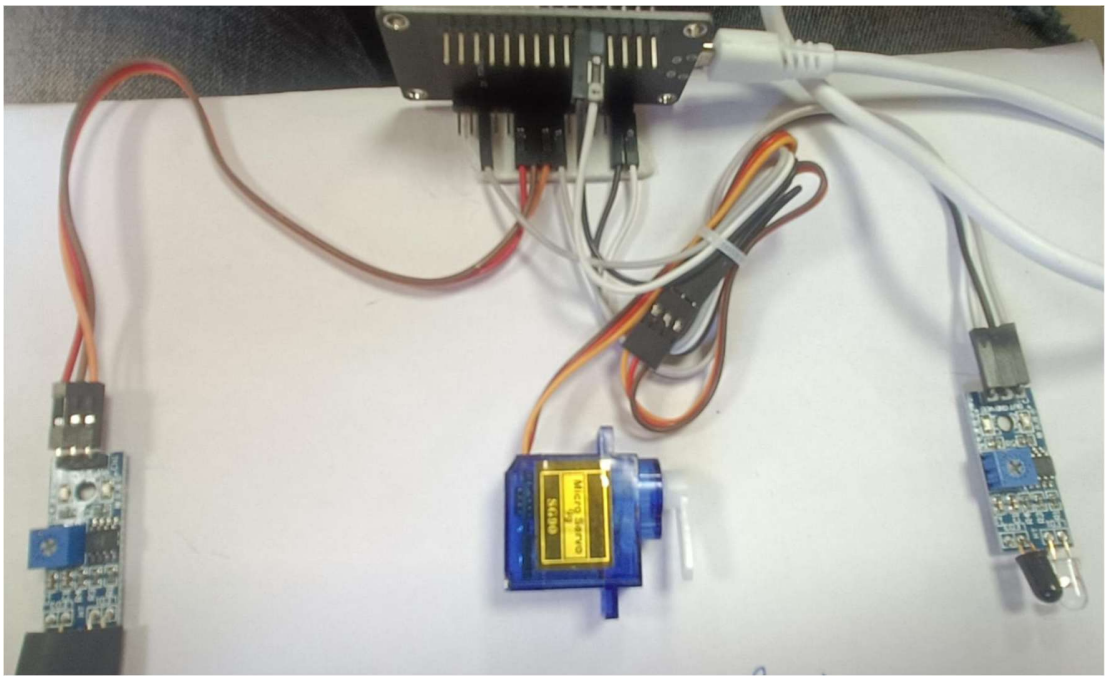
Role:

- Ensures proper signal transmission
- Connects modules to Arduino

## 4. CIRCUIT DIAGRAM

The circuit diagram represents the interconnection of all system components with the Arduino
Uno. The RFID module communicates with Arduino using SPI protocol, while the LCD uses I2C communication to minimize wiring complexity. IR sensors are connected to digital input pins to detect vehicle movement. The servo motor is connected to a PWM pin for precise control of gate movement.

When an RFID card is scanned, the Arduino processes the data and decides whether to grant access. Based on the decision, the servo motor operates the gate, and the LCD updates the system status accordingly.



## 5. CIRCUIT CONNECTIONS

### 1. RFID Module Connec ons (RC522)

RFID Pin Arduino Pin

SDA D10

SCK D13

RFID Pin Arduino Pin

MOSI D11 MISO

D12

RST D9

GND GND

3.3V 3.3V

### 2. IR Sensor Connec ons

Sensor Arduino Pin

Entry Sensor D2

Exit Sensor D3

### 3. Servo Motor Connec ons

Wire Arduino Pin Red

(VCC) 5V

Brown (GND) GND

Orange D6

### 4. LCD (I2C) Connec ons LCD Pin

Arduino Pin

VCC 5V

GND GND

SDA A4

SCL A5

### Power Connec ons

- Arduino powered via USB or adapter
- All components share a common ground

## 6. WORKING FLOW

1. RFID module scans user card
2. Arduino receives and verifies ID

3. IR sensors detect vehicle movement
4. Servo motor controls gate opening/closing
5. LCD displays real- me system status

## 7. WORKING PRINCIPLE

The system operates based on three major stages: Input, Processing, and Output.

### Step-by-Step Opera on:

1. System ini alizes and LCD displays “Scan Your Card”
2. User scans RFID card at entry
3. Arduino verifies card authen city

### Case 1: Authorized & Slots Available

- Servo motor opens gate
- IR sensor detects entry
- Slot count decreases  LCD shows “Access Granted”

### Case 2: Unauthorized Access

- Entry denied
- LCD displays “Access Denied”

### Case 3: Parking Full

- Entry blocked
- LCD displays “Parking Full”

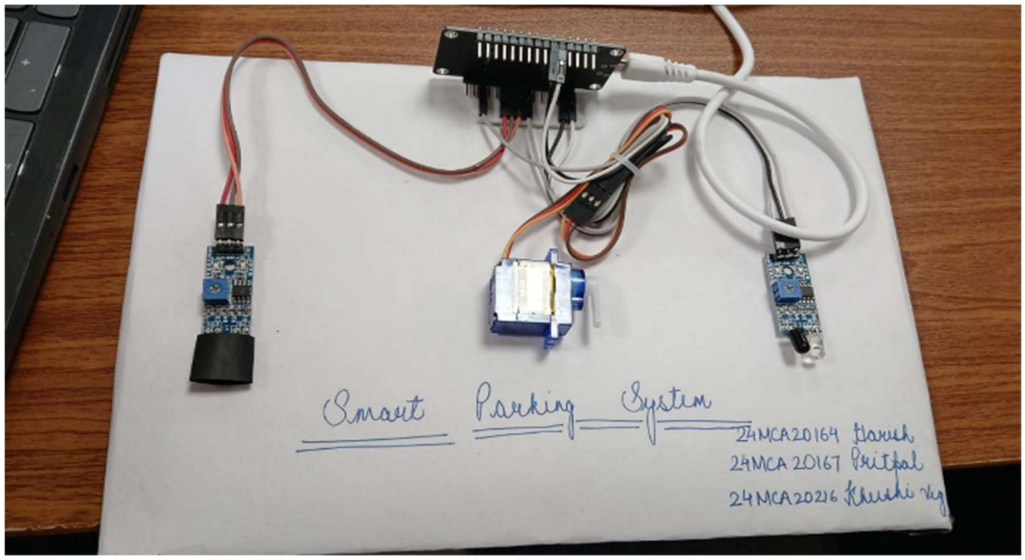
### Exit Process

- IR sensor detects vehicle exit
- Slot count increases
- LCD updates availability

The system continuously repeats this cycle, ensuring smooth and automated operation.

## 8. RESULT

The Smart Parking System using Arduino and RFID authentication was successfully designed and implemented. The system effectively automates vehicle entry and exit while ensuring security through RFID-based verification. It accurately monitors parking slot availability using IR sensors and provides real-time updates via the LCD display.

The servo motor efficiently controls the gate mechanism, ensuring smooth operation. The system proves to be reliable, cost-effective, and scalable for real-world applications such as shopping malls, residential complexes, office buildings, and smart city infrastructures.

This project demonstrates the practical implementation of embedded systems and IoT technologies in solving modern urban challenges and lays the foundation for future enhancements such as mobile app integration, cloud connectivity, and AI-based parking optimization.